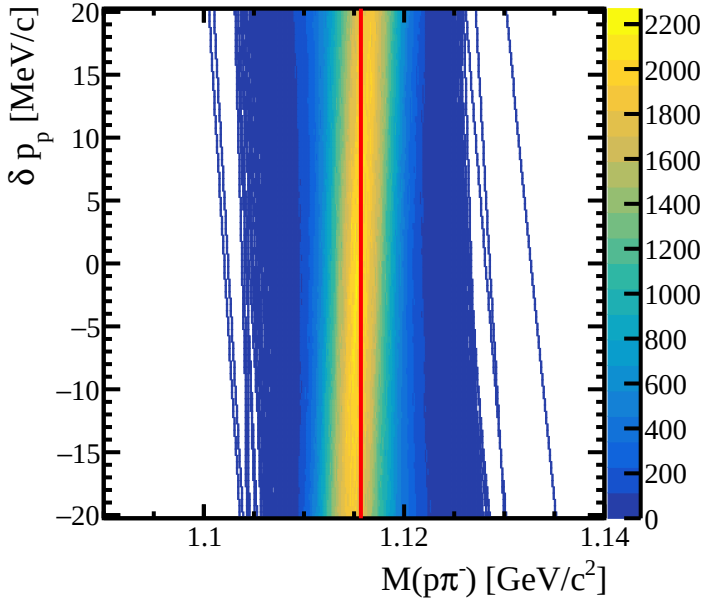
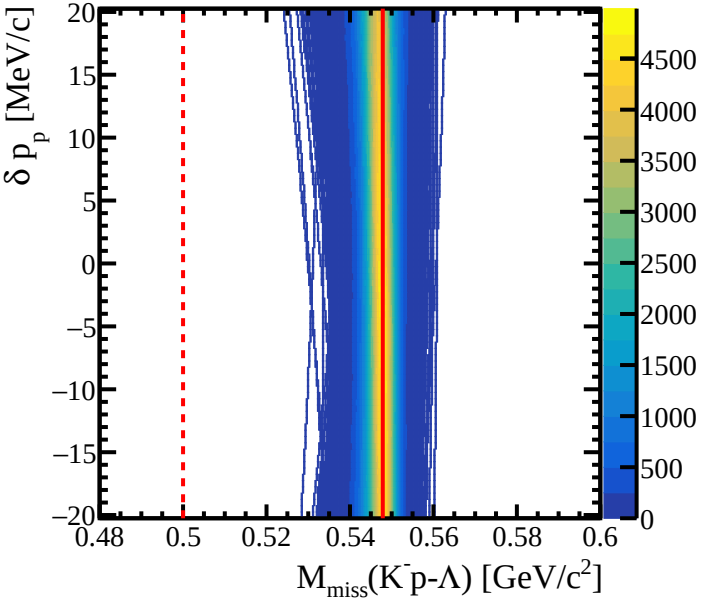
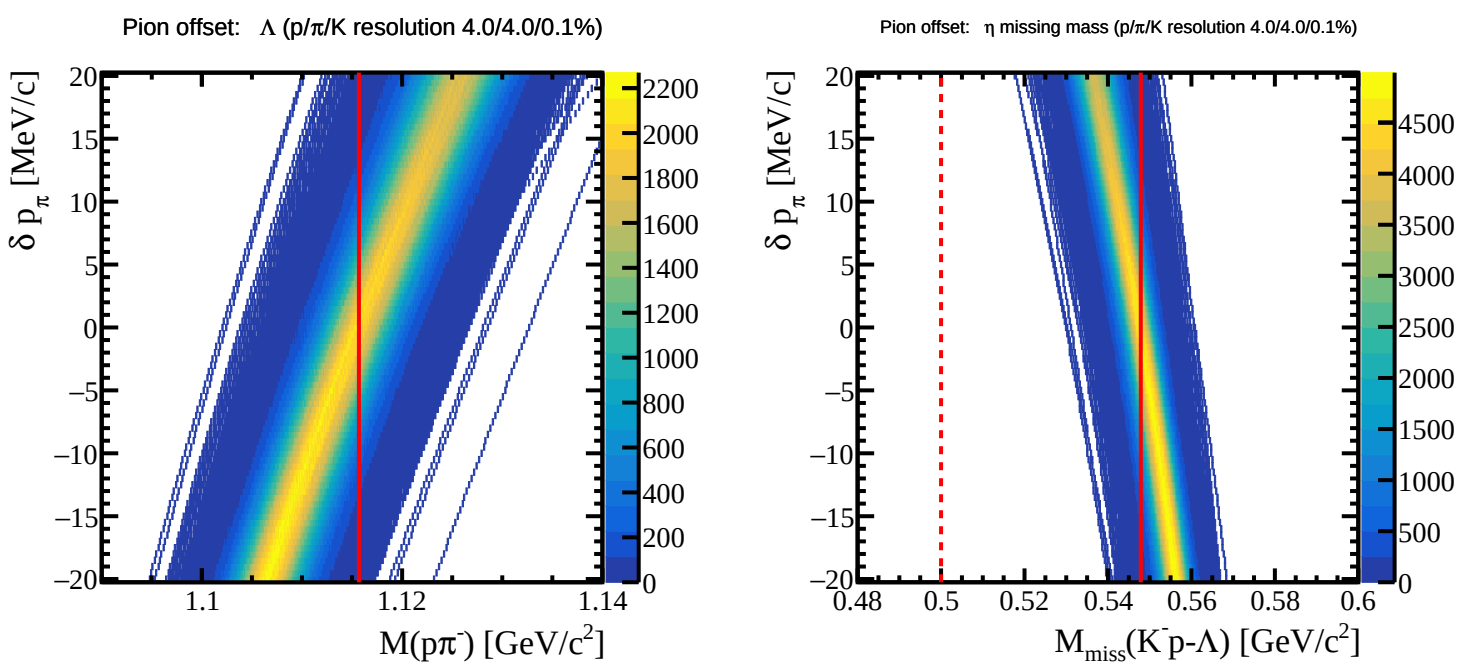
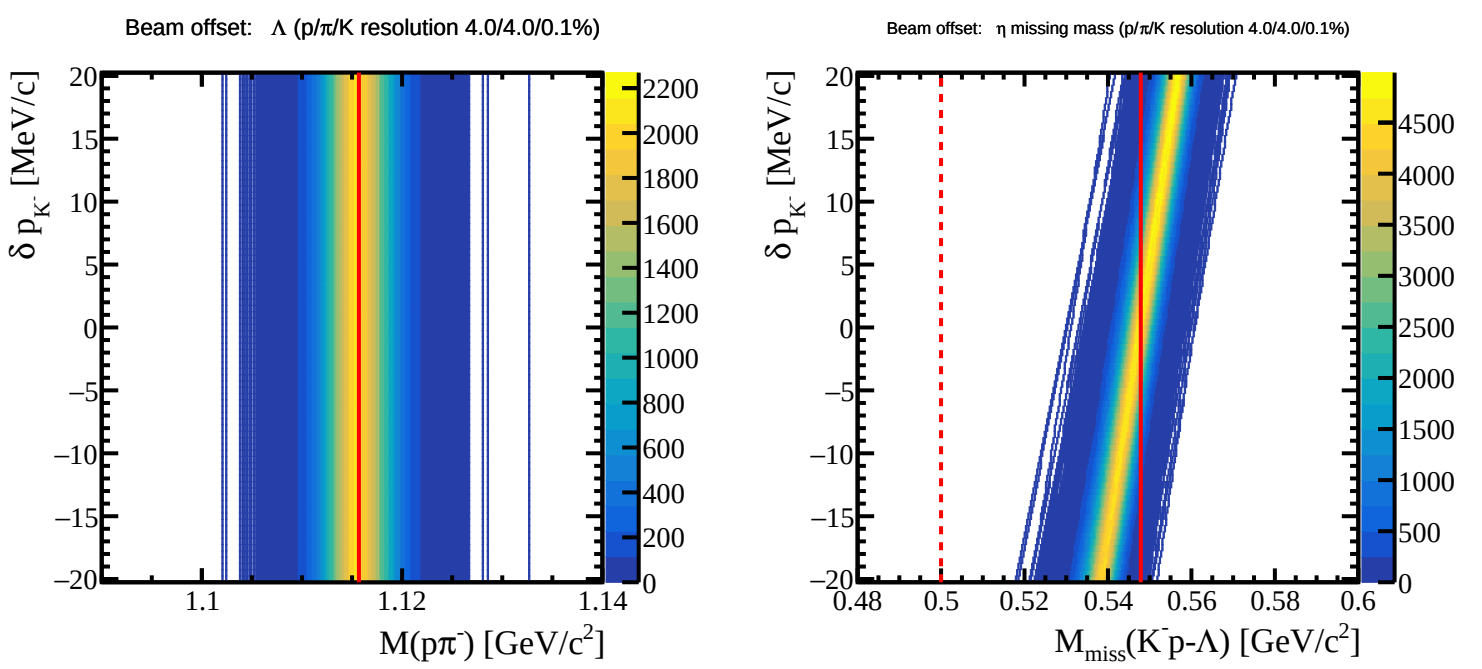
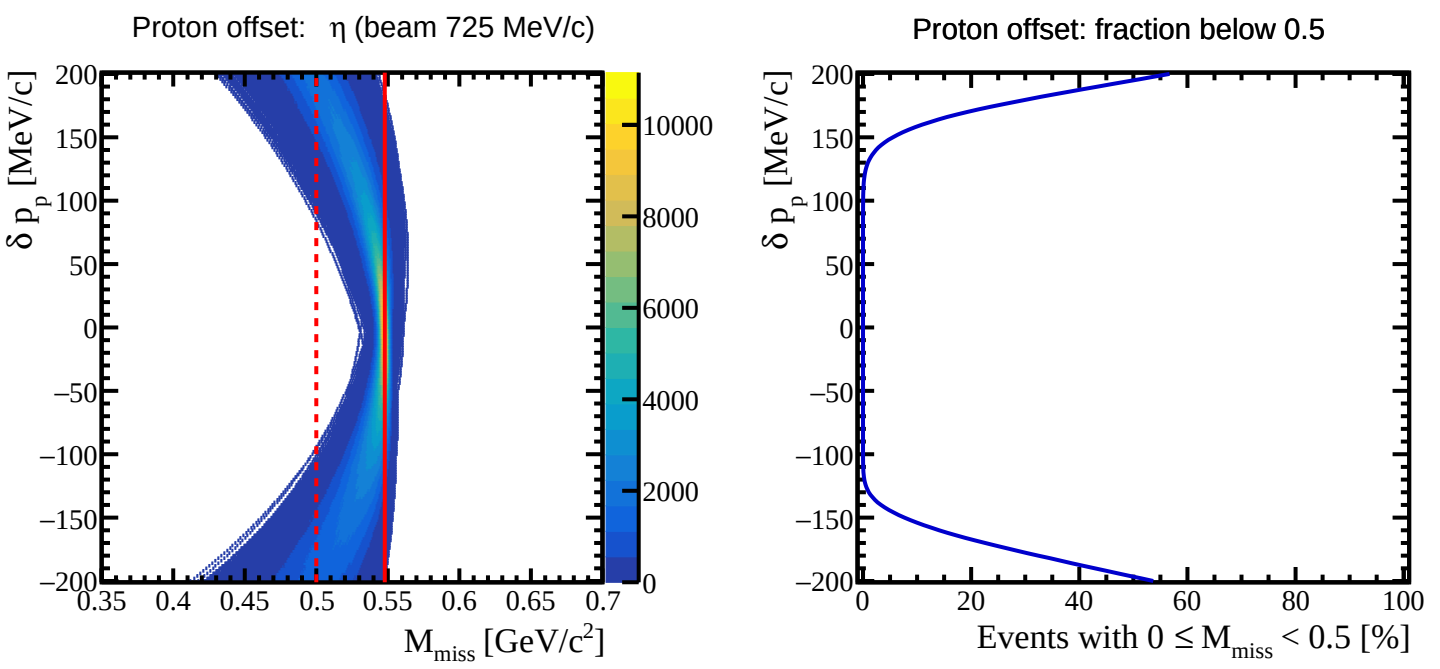
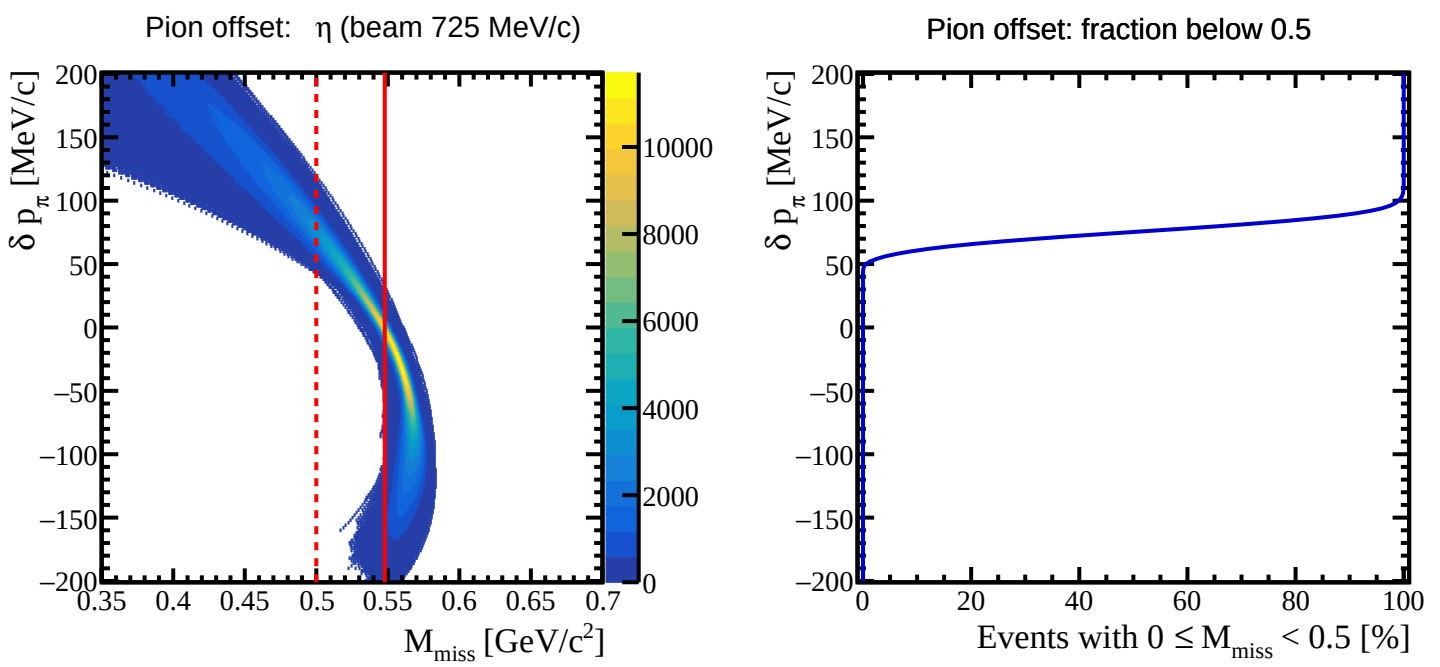

Proton offset: Λ (p/π/K resolution 4.0/4.0/0.1%)
δ p_p [MeV/c]
M(pπ⁻) [GeV/c²]

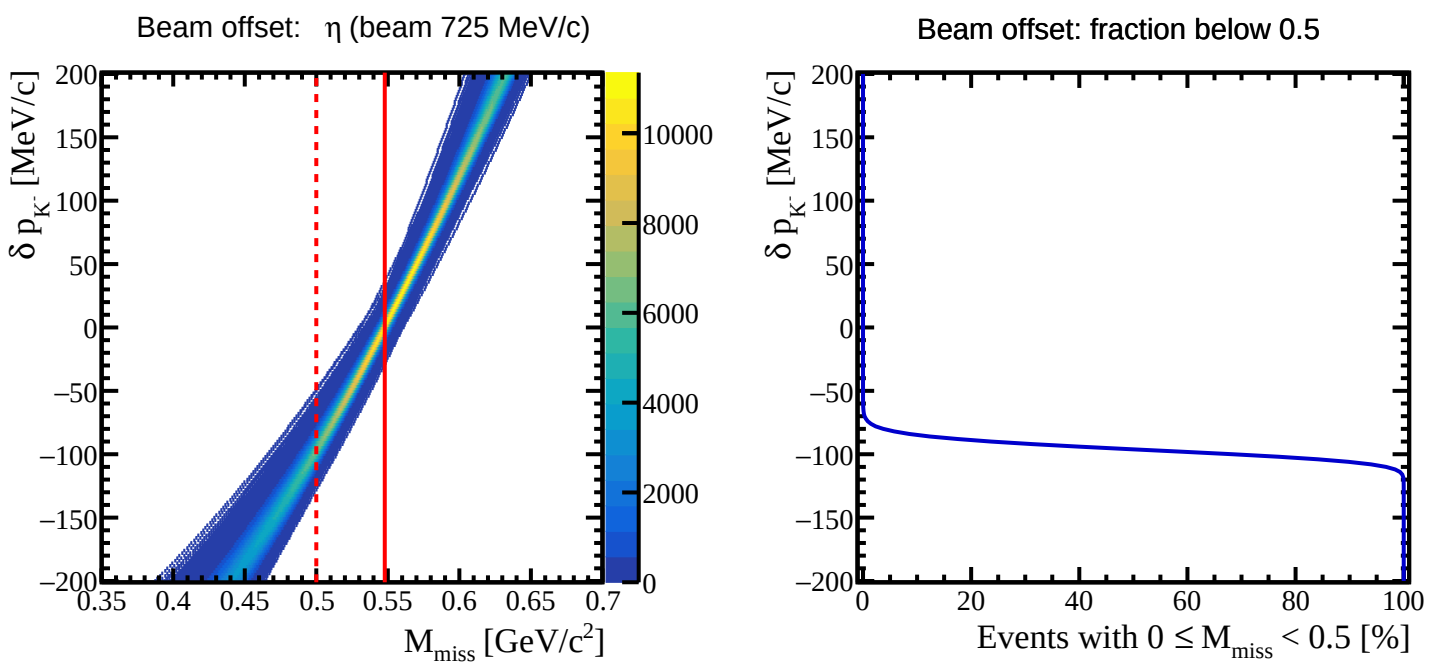

Proton offset: η missing mass (p/π/K resolution 4.0/4.0/0.1%)
δ p_p [MeV/c]
M_miss(K⁻p-Λ) [GeV/c²]

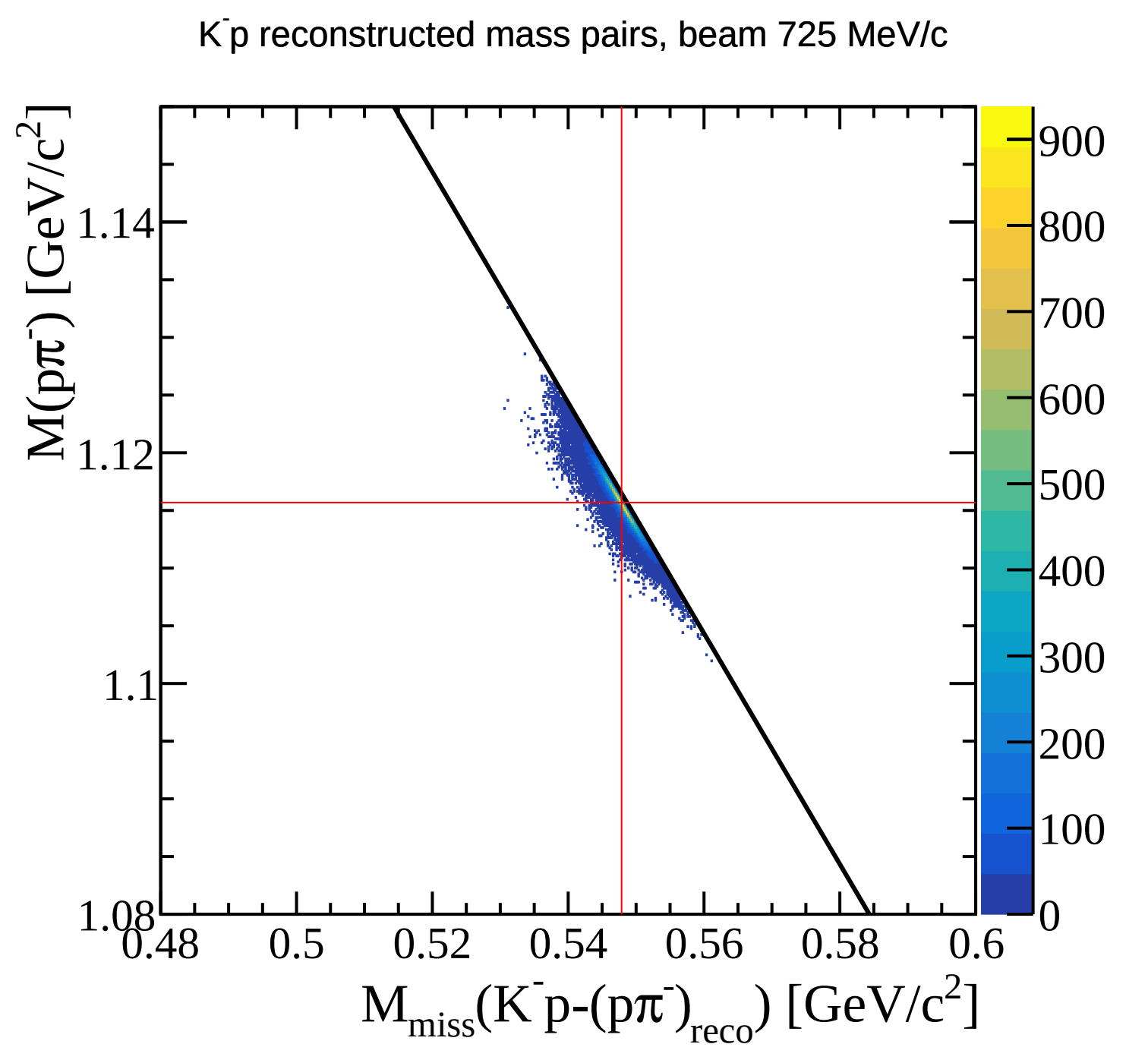

Pion offset: Λ (p/π/K resolution 4.0/4.0/0.1%)
δ p_π [MeV/c]
M(pπ⁻) [GeV/c²]
Pion offset: η missing mass (p/π/K resolution 4.0/4.0/0.1%)
δ p_π [MeV/c]
M_miss(K⁻p-Λ) [GeV/c²]

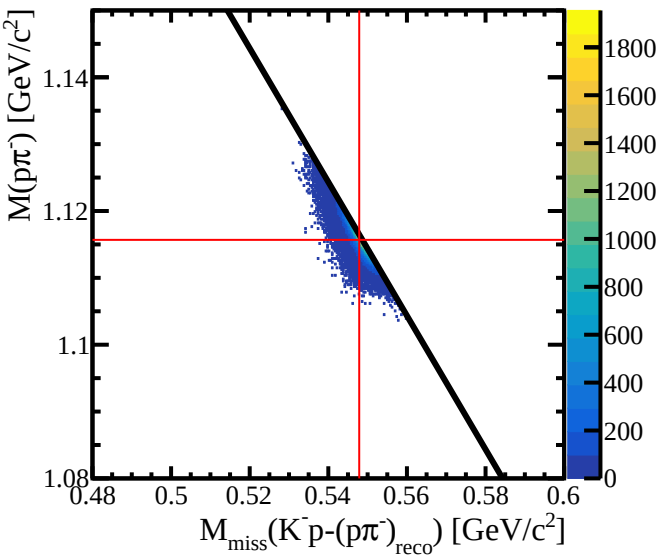

Beam offset: Λ (p/π/K resolution 4.0/4.0/0.1%)
δ p_K⁻ [MeV/c]
M(pπ⁻) [GeV/c²]
Beam offset: η missing mass (p/π/K resolution 4.0/4.0/0.1%)
δ p_K⁻ [MeV/c]
M_miss(K⁻p-Λ) [GeV/c²]

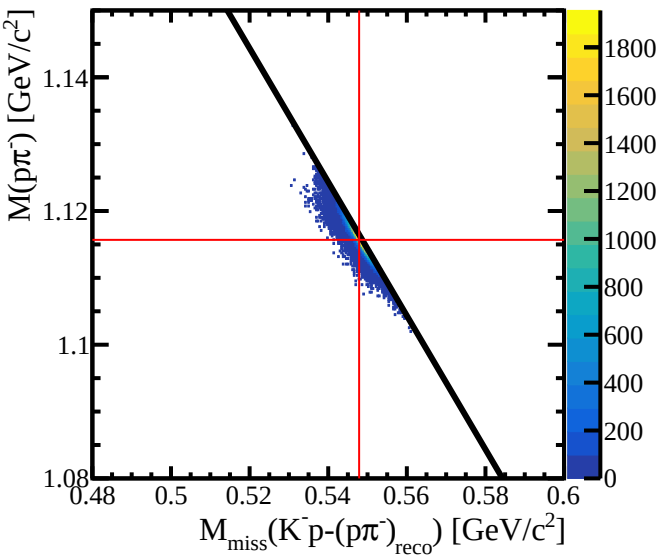

Proton offset: η (beam 725 MeV/c)
δ p_p [MeV/c]
200
150
100
50
0
−50
−100
−150
−200
0.35
0.4
0.45
0.5
0.55
0.6
0.65
0.7
M_miss [GeV/c²]
10000
8000
6000
4000
2000
0
Proton offset: fraction below 0.5
δ p_p [MeV/c]
0
20
40
60
80
100
Events with 0 ≤ M_miss < 0.5 [%]

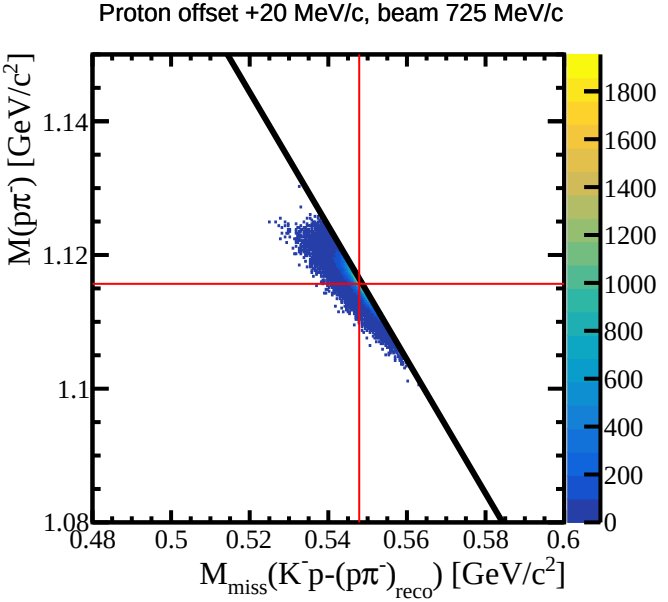

Pion offset: η (beam 725 MeV/c)
δ p_π [MeV/c]
M_miss [GeV/c²]
Pion offset: fraction below 0.5
δ p_π [MeV/c]
Events with 0 ≤ M_miss < 0.5 [%]

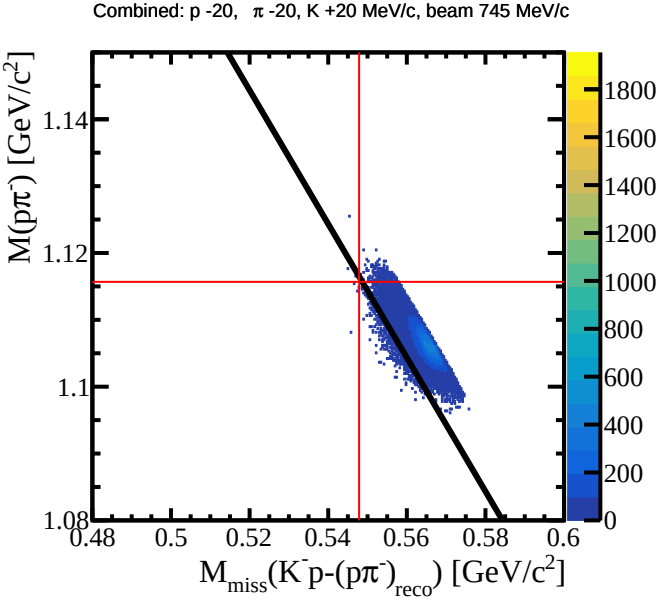

Beam offset: η (beam 725 MeV/c)
δ p_K⁻ [MeV/c]
M_miss [GeV/c²]
Beam offset: fraction below 0.5
Events with 0 ≤ M_miss < 0.5 [%]

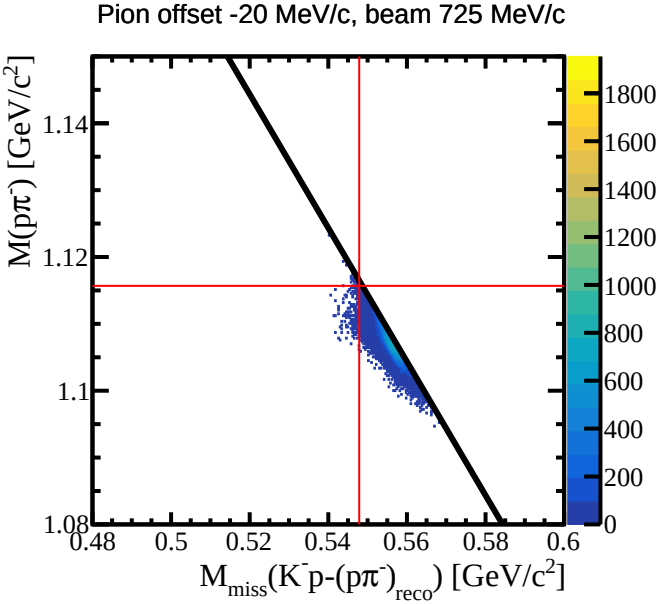

K⁻p reconstructed mass pairs, beam 725 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
900
800
700
600
500
400
300
200
100
0

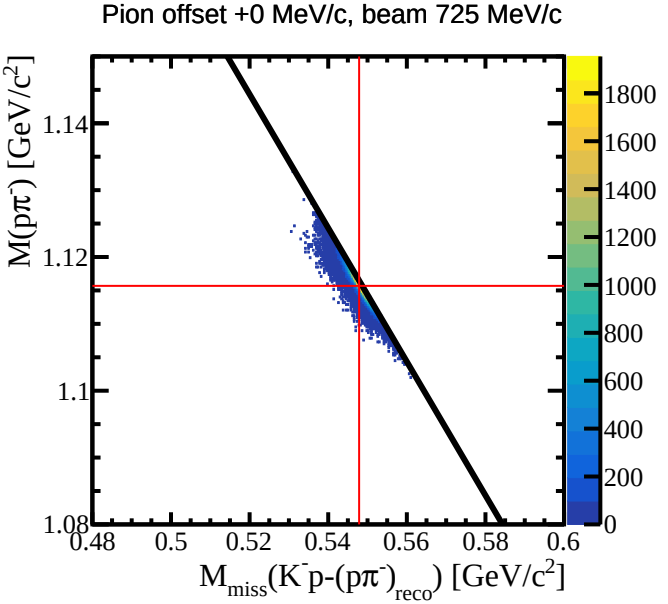

Proton offset -20 MeV/c, beam 725 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
1800
1600
1400
1200
1000
800
600
400
200
0

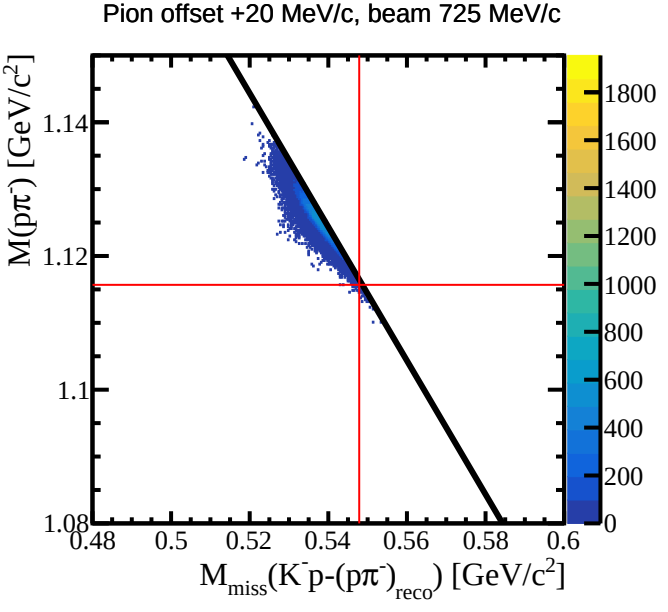

Proton offset +0 MeV/c, beam 725 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
1800
1600
1400
1200
1000
800
600
400
200
0

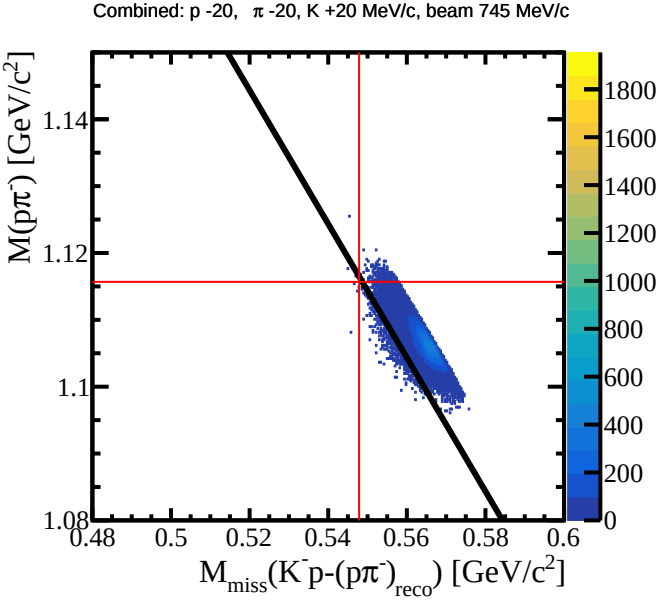

Proton offset +20 MeV/c, beam 725 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
1800
1600
1400
1200
1000
800
600
400
200
0

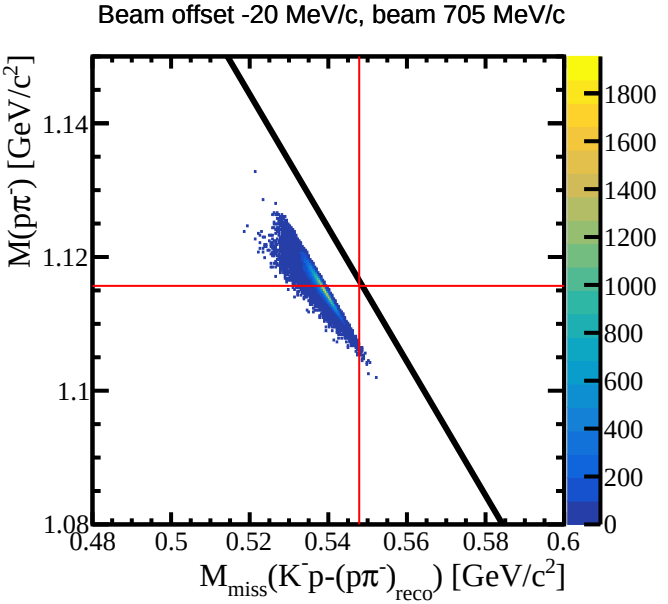

Combined: p -20, π -20, K +20 MeV/c, beam 745 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
1800
1600
1400
1200
1000
800
600
400
200
0

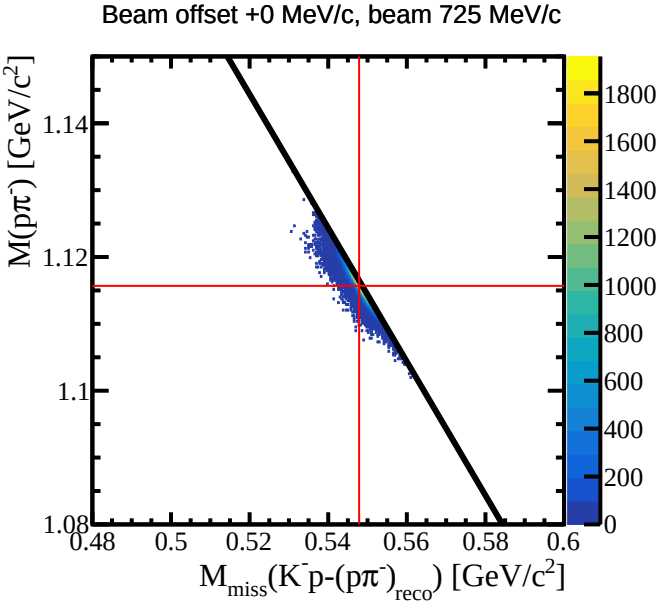

Pion offset -20 MeV/c, beam 725 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
1800
1600
1400
1200
1000
800
600
400
200
0

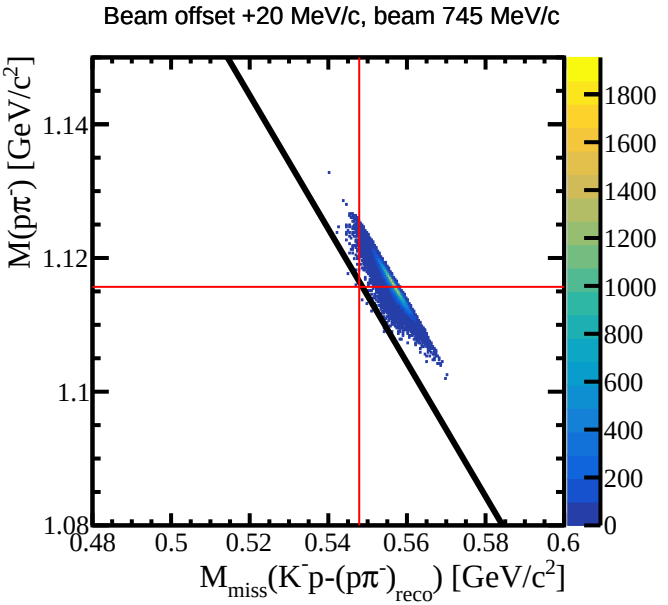

Pion offset +0 MeV/c, beam 725 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
1800
1600
1400
1200
1000
800
600
400
200
0

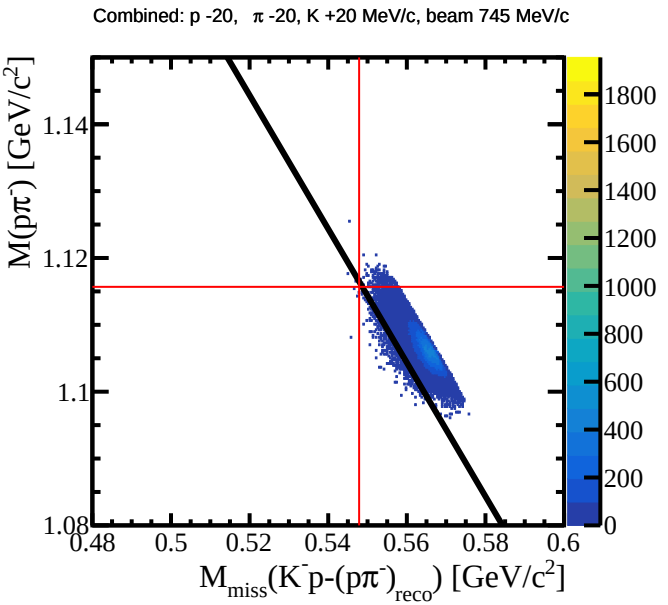

Pion offset +20 MeV/c, beam 725 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
1800
1600
1400
1200
1000
800
600
400
200
0

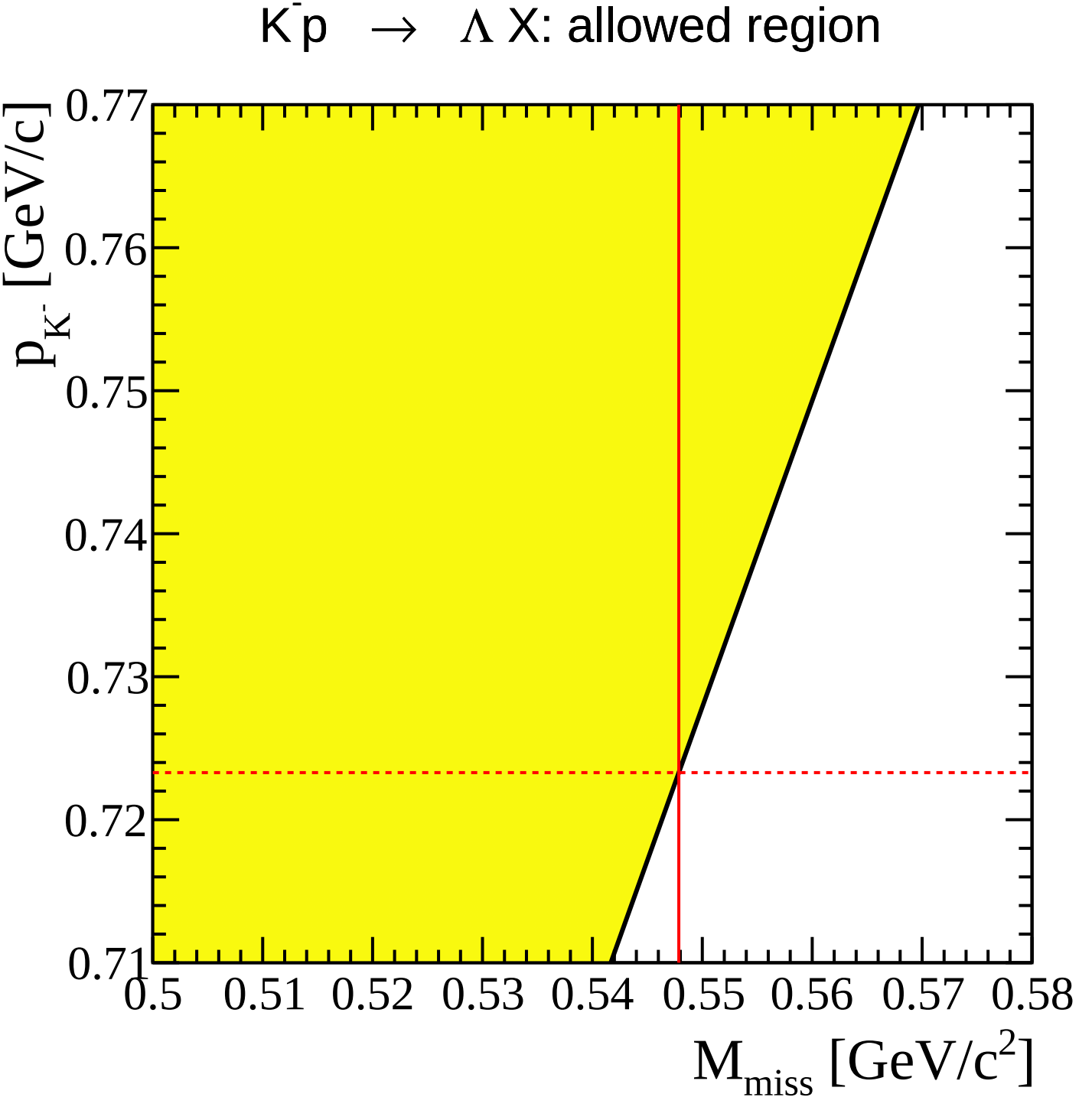

Combined: p -20, π -20, K +20 MeV/c, beam 745 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
1800
1600
1400
1200
1000
800
600
400
200
0

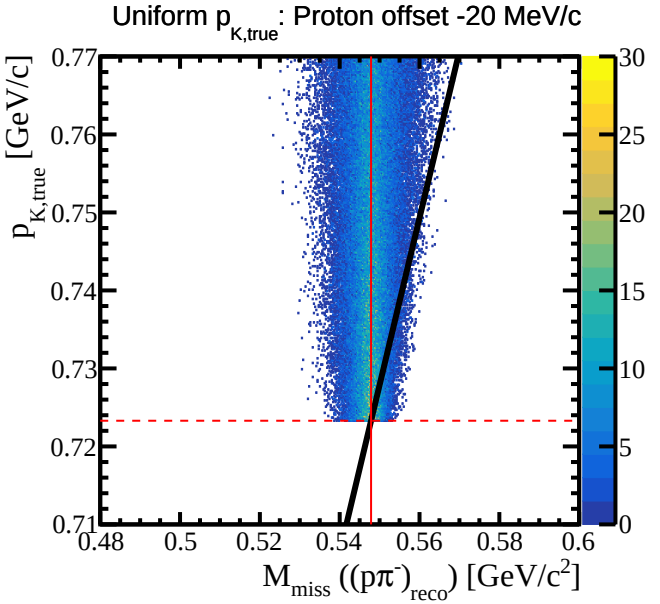

Beam offset -20 MeV/c, beam 705 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
1800
1600
1400
1200
1000
800
600
400
200
0

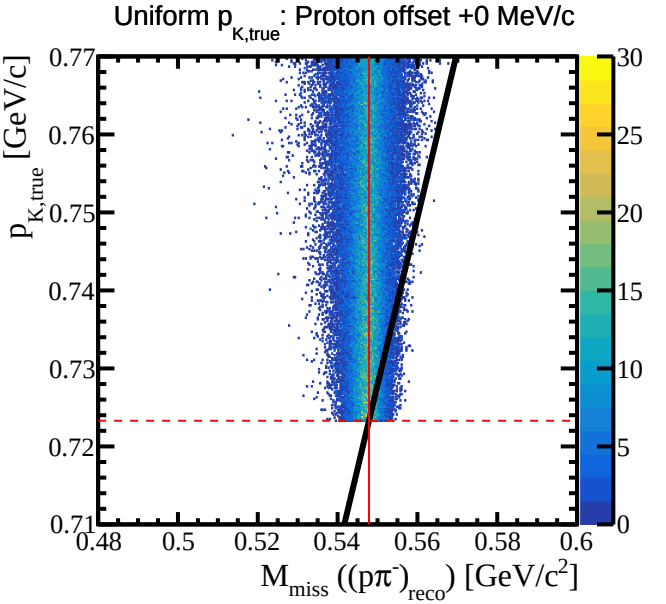

Beam offset +0 MeV/c, beam 725 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
1800
1600
1400
1200
1000
800
600
400
200
0

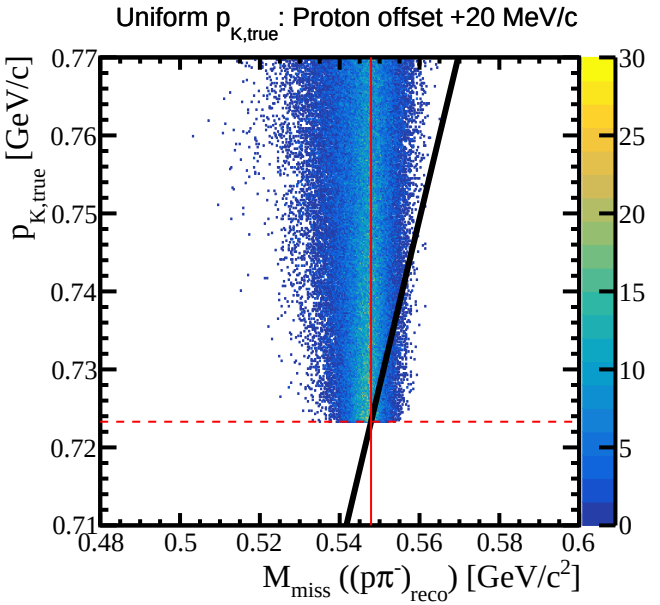

Beam offset +20 MeV/c, beam 745 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
1800
1600
1400
1200
1000
800
600
400
200
0

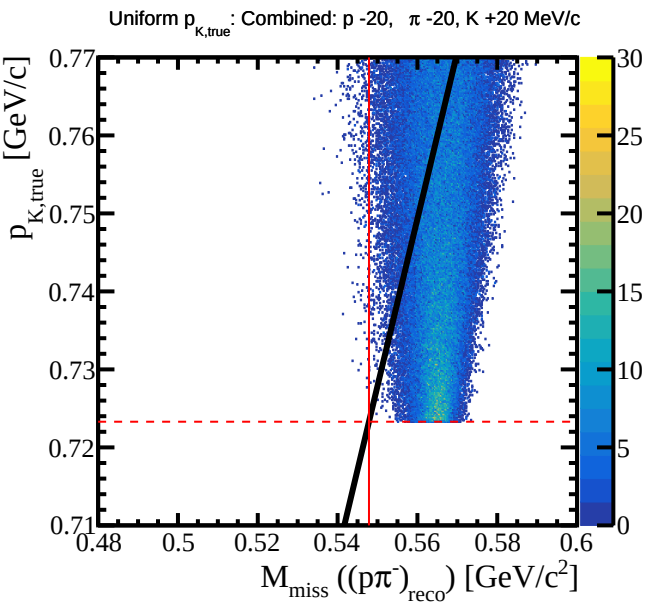

Combined: p -20, π -20, K +20 MeV/c, beam 745 MeV/c
M(pπ⁻) [GeV/c²]
1.14
1.12
1.1
1.08
0.48
0.5
0.52
0.54
0.56
0.58
0.6
M_miss(K⁻p-(pπ⁻)_reco) [GeV/c²]
1800
1600
1400
1200
1000
800
600
400
200
0

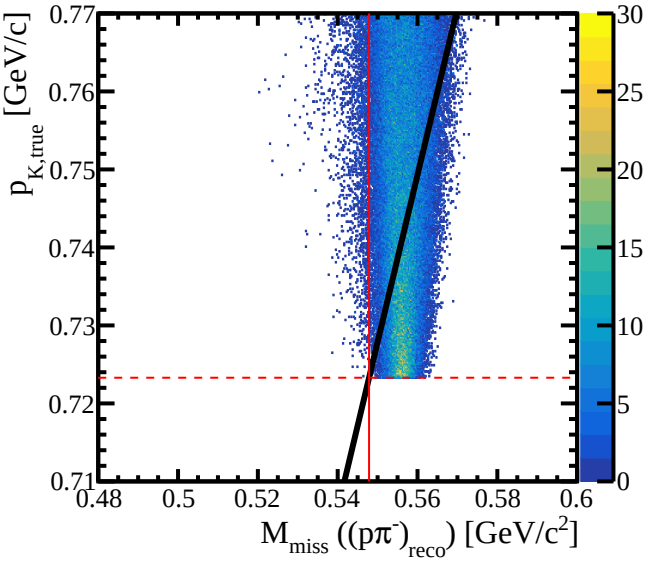

K⁻p → Λ X: allowed region
0.77
0.76
0.75
0.74
0.73
0.72
0.71
p_K⁻ [GeV/c]
0.5
0.51
0.52
0.53
0.54
0.55
0.56
0.57
0.58
M_miss [GeV/c²]

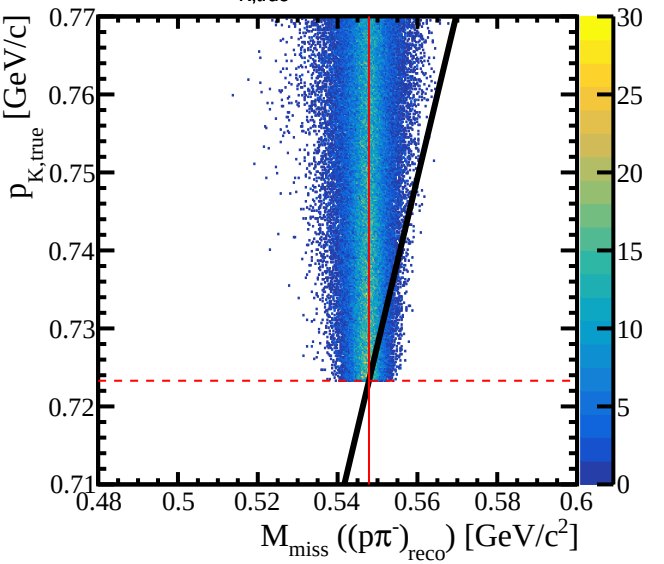

Uniform $p_{K,true}$: Proton offset -20 MeV/c
$p_{K,true}$ [GeV/c]
$M_{miss}((p\pi^-)_{reco})$ [GeV/c^2]

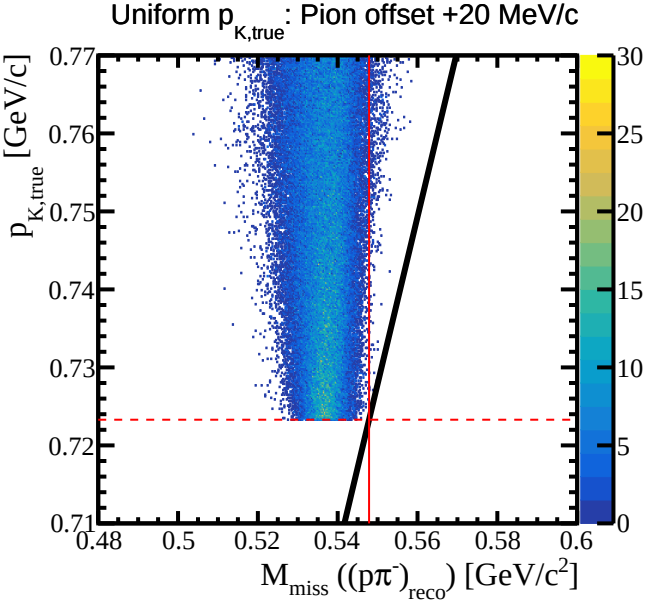

Uniform $p_{K,true}$: Proton offset +0 MeV/c
$p_{K,true}$ [GeV/c]
$M_{miss}((p\pi^-)_{reco})$ [GeV/c^2]

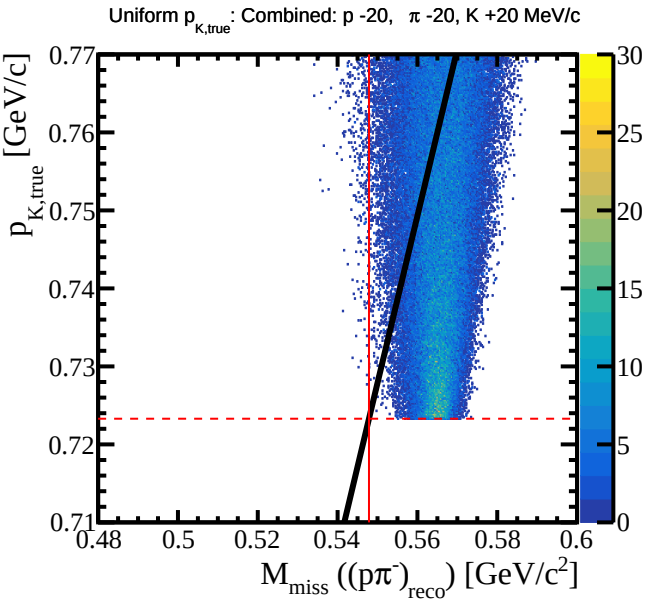

Uniform $p_{K,true}$: Proton offset +20 MeV/c
$p_{K,true}$ [GeV/c]
$M_{miss}((p\pi^-)_{reco})$ [GeV/c^2]

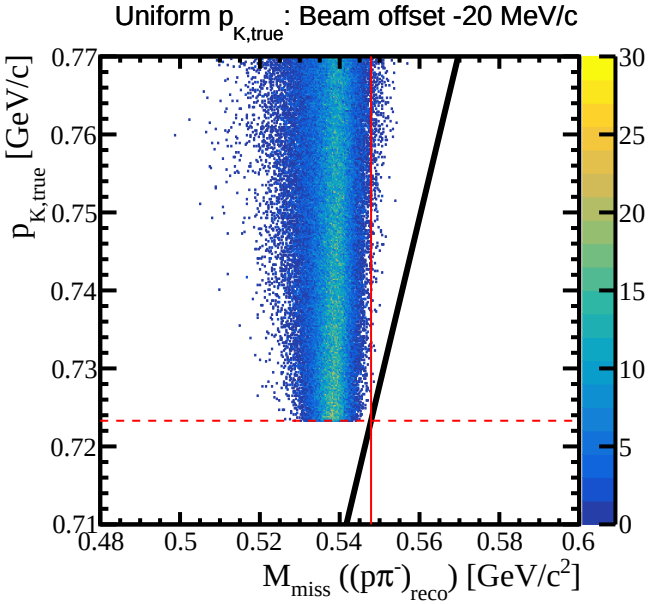

Uniform $p_{K,true}$: Combined: p -20, π -20, K +20 MeV/c
$p_{K,true}$ [GeV/c]
$M_{miss}((p\pi^-)_{reco})$ [GeV/c^2]

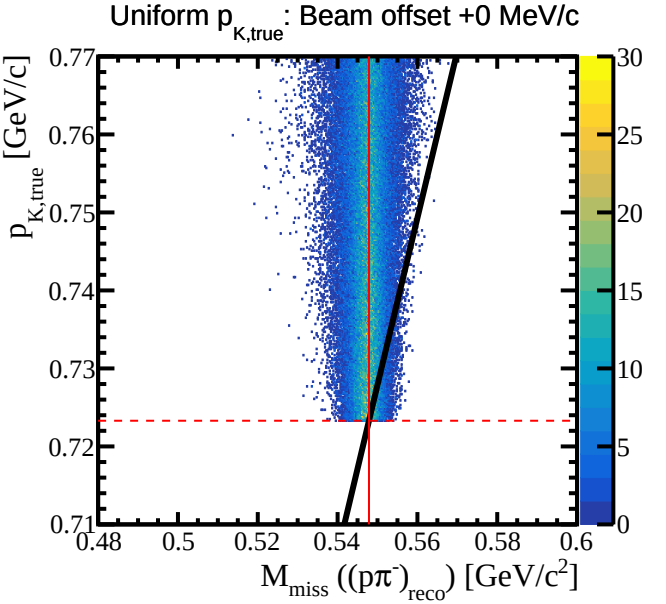

Uniform p_K,true: Pion offset -20 MeV/c
p_K,true [GeV/c]
M_miss ((pπ⁻)_reco) [GeV/c²]

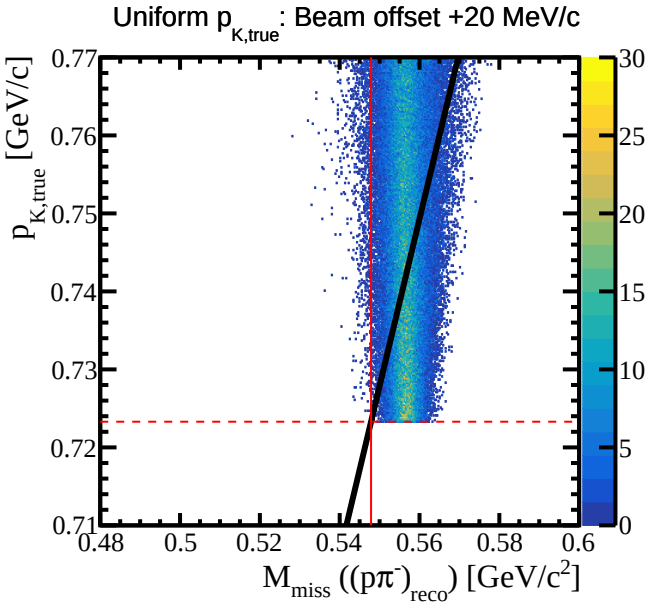

Uniform p_K,true: Pion offset +0 MeV/c
p_K,true [GeV/c]
M_miss ((pπ⁻)_reco) [GeV/c²]

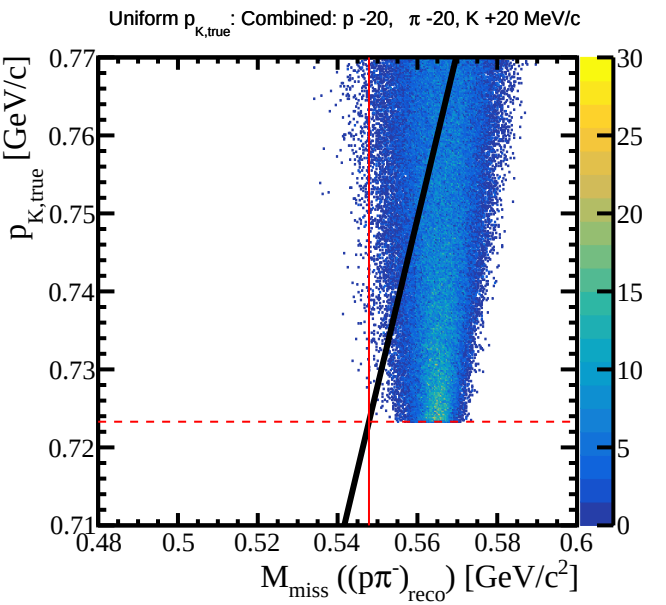

Uniform p_K,true: Pion offset +20 MeV/c
p_K,true [GeV/c]
M_miss ((pπ⁻)_reco) [GeV/c²]


Uniform p_K,true: Combined: p -20, π -20, K +20 MeV/c
p_K,true [GeV/c]
M_miss ((pπ⁻)_reco) [GeV/c²]


Uniform $p_{K,true}$: Beam offset -20 MeV/c
$p_{K,true}$ [GeV/c]
$M_{miss}((p\pi^-)_{reco})$ [GeV/c^2]


Uniform $p_{K,true}$: Beam offset +0 MeV/c
$p_{K,true}$ [GeV/c]
$M_{miss}((p\pi^-)_{reco})$ [GeV/c^2]


Uniform $p_{K,true}$: Beam offset +20 MeV/c
$p_{K,true}$ [GeV/c]
$M_{miss}((p\pi^-)_{reco})$ [GeV/c^2]


Uniform $p_{K,true}$: Combined: p -20, π -20, K +20 MeV/c
$p_{K,true}$ [GeV/c]
$M_{miss}((p\pi^-)_{reco})$ [GeV/c^2]